

Enzyme Kinetics

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- Effect of enzyme inhibitors

I. Introduction

Almost all chemical reactions that occur in biological systems are catalyzed by enzymes. Enzymes are highly specific biological catalysts of protein nature. To quantify an enzyme in a biological medium, it would be necessary to know its weight and molar mass in the pure state; however, this is only possible for a limited number of enzymes.

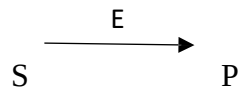
To overcome this practical problem, an enzyme can be characterized by studying the rate of the reaction it catalyzes.

I. General concepts

Kinetics is the study of the rates of chemical reactions. Enzyme kinetics aims to determine the rates of the reactions catalyzed by enzymes and to measure their affinity for substrates. It allows the identification and description of the mechanisms of biological reactions and enzyme mutations by studying their reaction rates. It also helps to understand how modifications in the enzyme's environment can affect its activity and functioning.

I.1. The different phases of an enzymatic reaction

Consider a reaction catalyzed by the enzyme E, which converts the substrate S into the product P:



- **Pre-steady-state phase:** when E is brought into contact with S, the formation of the E–S complex occurs rapidly (milliseconds).
- **Steady-state phase:** the enzyme is saturated with S, and the E–S complex reaches a constant maximal concentration.
- **Post-steady-state phase:** the concentration of S decreases significantly after a certain period of time.

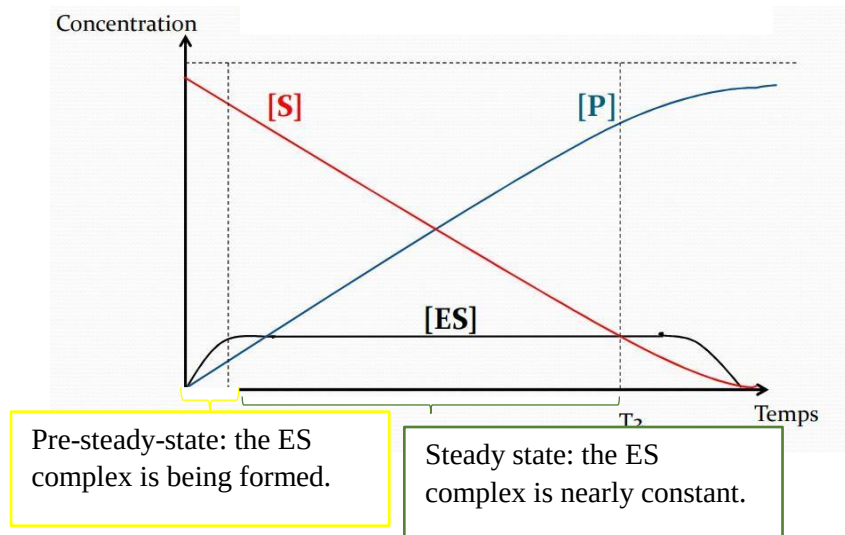


Figure 1 : The different phases of an enzymatic reaction

II. Initial Reaction Rate

The rate of a reaction is a measure of how quickly the concentrations of reactants and products change over time.

It is expressed as:

- The amount of substrate metabolized per unit time:

$$V = -d[S]/dt$$

- Or the amount of product formed per unit time:

$$V = d[P]/dt$$

- For given concentrations of [E] and [S], the amount of product P formed is measured as a function of time:

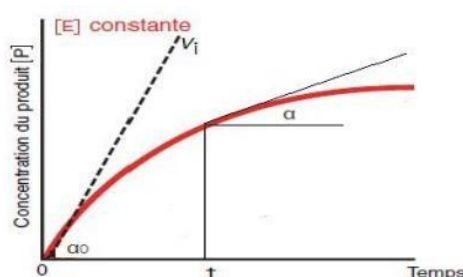


Figure 2 : Change in product P as a function of time

At time t , $V = \tan \alpha$

I.1. Factors Influencing Enzyme Kinetics

1. Temperature

- Each enzyme has an **optimal temperature** at which it shows maximum activity.
- The **maximum reaction rate (V_{max})** increases proportionally with temperature until it reaches a **maximum plateau (optimal temperature of activity)**. Beyond this plateau, the reaction rate decreases, indicating **denaturation of the enzymatic protein at high temperatures**.
- In humans, the **optimal temperature range** is generally **between 25–40 °C**.

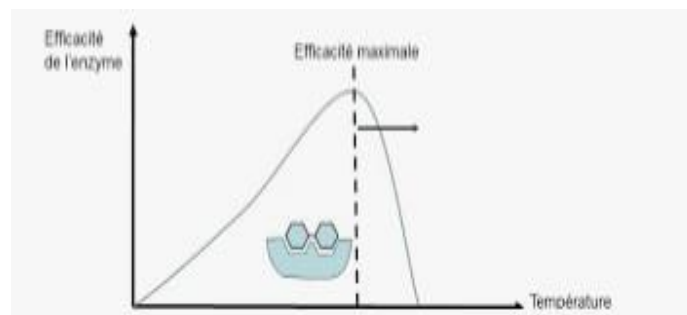


Figure 2 : Effect of Temperature on Enzymatic Activity

2. pH of the Medium

The enzyme kinetics obtained by varying the pH shows a similar pattern to that obtained when varying temperature, except that the optimal pH depends on the structure of the enzyme (which is composed of amino acids). Thus, in humans, the optimal pH range may be neutral, acidic, or alkaline, depending on the enzyme considered.

Example: Lactate dehydrogenase (LDH) shows maximum activity at a neutral pH.

3. Enzyme Concentration

When the **substrate concentration [S]** is constant, the **reaction rate increases proportionally with the increase in enzyme concentration**. This occurs because a greater number of enzyme molecules provides **more active sites available for the substrate**, leading to a higher reaction rate.

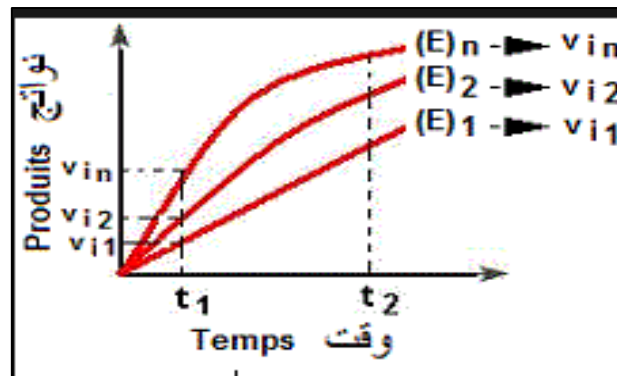


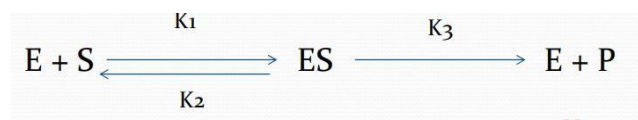
Figure 3 : Effect of the concentration on enzymes activity

4. Substrate Concentration

The rate of an enzymatic reaction increases proportionally with the increase in substrate concentration (S) until it reaches a maximum value, called V_{max} , which corresponds to the saturation of the enzymes. At this point, all the active sites of the enzymes are occupied by substrate molecules, and the reaction rate can no longer increase further.

V. Study of Michaelis–Menten kinetics

During an enzymatic reaction, there are two important steps. The first is the formation of an enzyme–substrate complex, then during the second step, the product is formed and the enzyme is released. According to the reaction:



with k = rate constant.

- k_1 = rate constant for the formation of the ES complex
- k_2 = rate constant for the dissociation of the ES complex
- k_3 = rate constant for the formation of P

Conditions of the Michaelis–Menten equation:

Saturating substrate concentration: all the enzyme is in the ES form.

The Michaelis curve is hyperbolic, and the kinetic characteristics are:

- The maximum velocity (V_m) is reached when all the enzyme is saturated (ES complex).
- The Michaelis constant K_m is equal to the substrate concentration when $V = \frac{1}{2} V_m$.
- It represents the inverse of the enzyme's affinity for the substrate.

$$K_m = K_2 + K_3 / K_1$$

Equation of Michaelis-Menten :

$$V_i = V_m [S] / K_m + [S]$$

With:

- v_i : initial velocity (i.e., in the absence of product) of the enzymatic reaction for a given substrate concentration $[S]$ (in $\mu\text{mol/min}$);
- v_{max} : maximum initial velocity measured for a saturating substrate concentration (in $\mu\text{mol/min}$);
- $[S]$: substrate concentration (in mol/L);
- K_M : Michaelis constant specific to the enzyme. It is the substrate concentration at which the initial reaction velocity is equal to half of the maximum initial velocity $\frac{v_{\text{max}}}{2}$ (in mol/L)

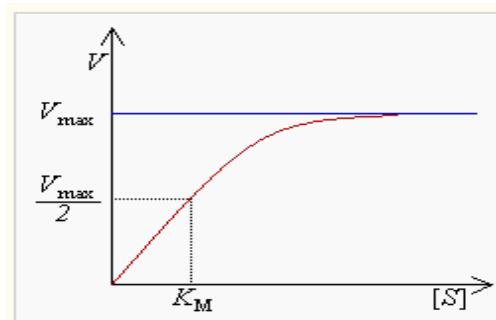


Figure 4 : Representation of Michaelis-Menten

Note:

K_m reflects the affinity of the enzyme for its substrate. Affinity represents how easily a substrate binds to an enzyme. Therefore, the higher the K_m , the more the ES complex tends to dissociate, meaning the enzyme has **lower affinity** for the substrate. K_m is thus **inversely proportional** to the enzyme's affinity for its substrate.

VI. Lineweaver–Burk Representation

- A hyperbola is difficult to draw manually.
- Errors in estimating V_{max} are possible.
- To simplify the graphical representation of the Michaelis–Menten equation, the hyperbola is transformed into a straight line.

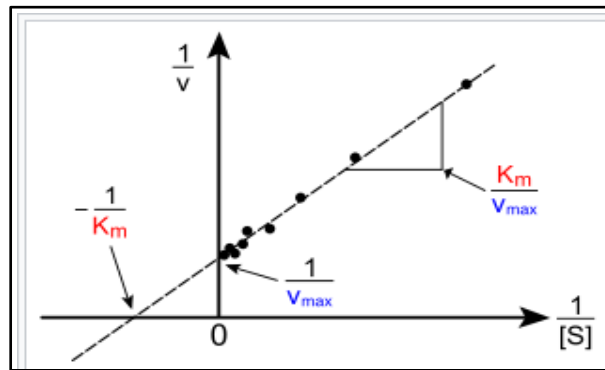


Figure 5 : Representation of Lineweaver-Burk The Lineweaver–Burk equation is the double reciprocal form of the Michaelis–Menten equation, expressed as:

$$1/V = \frac{K_m}{V_{\max}} \times \frac{1}{[S]} + \frac{1}{V_{\max}}$$

This equation is a straight-line equation of the form $Y = aX + b$, or $1/V = f(1/[S])$. It involves plotting **1/V versus 1/[S]**.

- The resulting **line** intercepts the y-axis at $1/V_{\max}$ and the x-axis at $-1/K_m$.
- This representation is easier to read. Specifically:
 - $V_{\max}$ is determined from the **intersection of the line with the y-axis**.
 - K_m is determined from the **intersection of the line with the x-axis**.

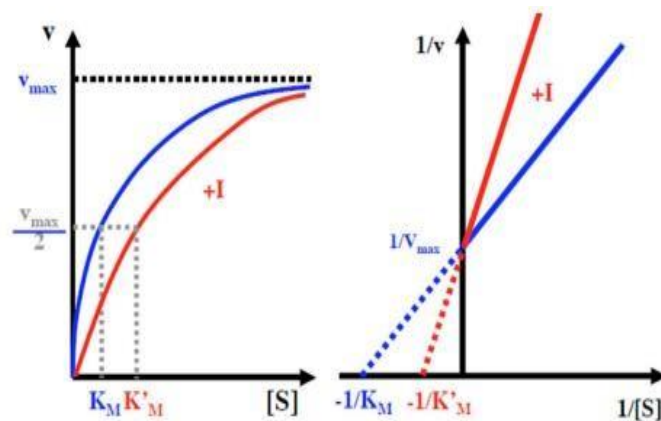
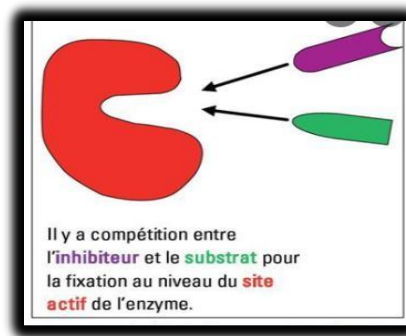
V. Effect of Enzyme Inhibitors

Inhibitors are substances that inactivate enzymes. There are two main types of inhibition:

- **Reversible enzyme inhibition:** enzymatic activity can be restored by removing the inhibitor.
- **Irreversible enzyme inhibition:** the inhibitor binds covalently to the enzyme, inactivating it permanently.

1. Competitive Inhibitors

A competitive inhibitor competes with the substrate for the active site, forming the enzyme–inhibitor complex [E–I]. This type of inhibition can be reversed by increasing the substrate concentration [S].

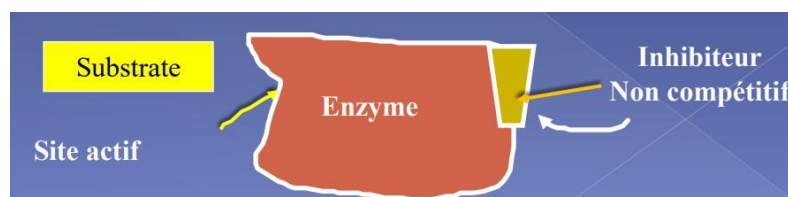


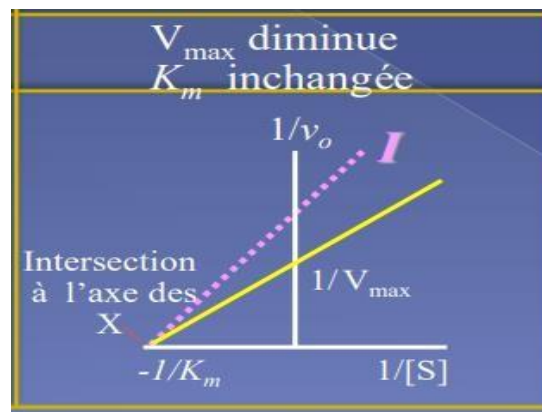
$$\frac{1}{v} = \frac{K_M \left(1 + \frac{[I]}{K_i} \right)}{V_{\max}} \cdot \frac{1}{[S]} + \frac{1}{V_{\max}}$$

Kinetically, $V_{\max}$ remains unchanged, while K_M is increased.

2. Non-competitive inhibitors

A non-competitive inhibitor binds to both the enzyme (E) and the enzyme-substrate complex (ES) at a site different from the active site, so there is no competition between the substrate and the inhibitor. This results in a decrease in the maximum velocity ($V_{\max}$) without any change in K_M , since the inhibitor does not compete with the substrate for the active site.

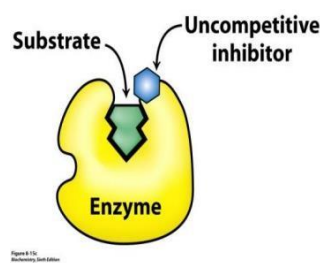




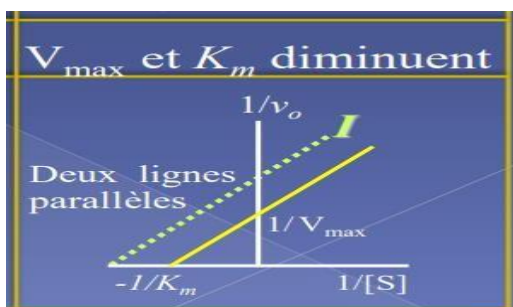
This inhibitor shifts the equation; K_m is not changed. Part of the ES complex remains blocked as ESI, so V_{max} is decreased.

3. Uncompetitive inhibitors

An uncompetitive inhibitor never binds to the free enzyme but only to the ES complex, preventing product formation. Binding of the substrate to the enzyme induces a change in the enzyme, exposing a binding site for the inhibitor. In turn, the inhibitor alters the conformation of the enzyme's active site, thereby blocking the reaction.



The inhibitor binds to the ES complex, forming ESI.



This inhibitor shifts the equation and lowers K_m . Part of the ES complex remains blocked as ESI, so V_{max} is decreased

